# AUTO SCALING FOR HIGH AVAILABILITY

---

**WHAT IS AUTOSCALING**

1. it is a service which is provided by aws to autoscale up and scale down the instances based on the demand for your application.
2. autoscaling helps you save cost by cutting down the number of EC2 instances when not needed and scaling out to add the instances when it is required.
3. it works on the scaling policies that we can attach to the autoscaling group.
4. When the demand for your application is increases, autoscaling will automatically add the instances to handle the load.
5. Conversely, when the demand for your application is decreases, autoscaling will automatically remove unnecessary instances to save the cost.
6. It improves high availability and performance.
7. It ensures your application is always available and responsive.

**TYPES OF AUTOSCALING**

1. **Vertical Scaling**
    1. With vertical autoscaling, you can add or remove CPU, memory, storage, or other resources to an instance to increase or decrease its capacity.
    2. In other words, it involves adding more resources to an existing instance, rather than adding more ec2 instances.
2. **Horizontal Scaling**
    1. Horizontal scaling involves adding more instances to a system to handle increased demand or workload.
    2. Horizontal scaling is often used for applications or services that experience high traffic or workload spikes, where additional computing resources are needed to handle the increased demand.
    3. Horizontal scaling can be more cost-effective than vertical scaling (scaling up), as adding additional smaller instances is often cheaper than adding more resources to a single larger instance.

---

**TYPES OF SCALING IN AUTOSCALING**

1. Manual scaling
2. Dynamic scaling (Live-data)
    1. Target Tracking Policy
    2. Simple Scaling Policy
    3. Step Scaling Policy
3. Scheduled Scaling (Time-based)
4. Predictive Scaling (Future-based)

---

**AUTOSCALING CAPACITIES**

1. **Desired**
   This is the number of instances you WANT to run normally

- Auto Scaling will always try to maintain this number
- If an instance crashes → it launches a new one
- If scaling policy triggers → this value changes automatically
- Desired = 3 → AWS will always keep 3 EC2 instances running

2. **Minimum**
   This is the LOWEST limit

- Auto Scaling will never go below this number
- Even if traffic is very low
- Min = 2 → Even at night/low traffic, 2 instances will always run

3. **Maximum**
   This is the UPPER limit

- Auto Scaling will never exceed this number
- Helps control cost and resource usage
- Max = 6 → Even during heavy traffic, it will not go beyond 6 instances

---

**SCALING CALCULATIONS CONCEPT**

- Minimum = 1
- Desired = 2
- Maximum = 5

**IF I ADD THE INCREASE POLICY ON THE AUTOSCALING GROUP**

**If CPU ≥ 60% → Add 2 instances:**

- Current instances = 2
- Policy says → +2 instances

New instances = 2 + 2 = 4

So now 4 instances will be running

**Let's say CPU is still high:**

- Current = 4
- Add 2 → 4 + 2 = 6

BUT

Maximum Capacity = 5

So AWS will NOT go beyond 5

Final = 5 instances (capped by max)

**Final Instances = Desired + Scaling Adjustment (but within Min & Max limits)**

---

**COOLDOWN PERIOD**

cooldown period simply means when the instances is created or terminated the autoscaling take time to perform the next scaling action

---

**WARM-UP PERIOD**

Warm-up period is the time given for newly launched instances to become fully ready before their metrics are considered for further scaling decisions.

---

**PRACTICAL ON AUTOSCALING**

**1.Create A Launch Template**

1. name ----------> webserver-LT
2. tags ----------> webserver
3. instance_type --> t3. Micro
4. vpc ----------> default

5. sg ----------> ssh/http/https
6. create a instance

**2.Create A Autoscaling Group**

1. name ----------> ASG
2. LT ----------> webserver-LT
3. Vpc ----------> default
4. AZ ----------> 1a/1b/1c
5. Capacity
    a. Desired ---- 1
    b. Minimum --1
    c. Maximum – 5
6. Automatic scaling ----> target tracking scaling
    a. Metric type – average cpu utilization
    b. Target value – 50%
7. Create a autoscaling group

**3.Access The Instance & Add the stress**

1. apt install stress -y
2. stress --cpu 4 –timeout 300s

---

**TASK**

1.create a autoscaling on windows instances

**Note:**

1. create a file and add the a.bat in it and save these as a.bat with all files
2. click on these file multiple times (open this file in multiple terminals)

---